



NATIONAL UNIVERSITY OF MODERN LANGUAGES

SOFTWARE QUALITY ENGINEERING LAB REPORT – 4

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PROGRAM: BSSE (6th Semester)

COURSE: SOFTWARE QUALITY ENGINEERING

SUBMITTED TO: Mam Rubab

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Project: RAG-Based AI Voice Agent for Businesses

Objective:

To map the quality requirements of our final year project (a voice agent for businesses) to two established software quality models and justify the selection of these models.

1. McCall’s Quality Model Mapping

McCall’s model breaks software quality into factors like operation, maintenance, and adaptability. For our voice agent, the mapping is as follows:

Quality Factor	Application to Voice Agent
Correctness	The agent provides accurate answers to user queries.
Reliability	The system works consistently without crashes or downtime.
Efficiency	Responses are delivered quickly to avoid user frustration.
Integrity	Business and customer data is handled securely and confidentially.
Usability	The interface is simple and intuitive for staff and customers.
Maintainability	Updates to the knowledge base and system improvements are easy to implement.
Flexibility	The system can adapt to different business domains or languages.

2. FURPS Model Mapping

FURPS focuses on functionality, usability, reliability, performance, and supportability. For the voice agent, the mapping is:

Category	Application to Voice Agent
Functionality	Understands voice commands, retrieves information, and supports conversations.
Usability	Simple to use, requires minimal training, and provides clear prompts.
Reliability	Handles multiple queries simultaneously without errors.
Performance	Provides fast responses even with a high number of users.
Supportability	The knowledge base can be updated easily, and logs help monitor performance.

3. Justification for Selected Models

- McCall's Model:**
 It is well-suited for evaluating operational performance and maintenance. Our voice agent needs to work reliably for businesses and be adaptable over time. McCall's factors like correctness, reliability, and maintainability directly relate to these requirements.
- FURPS Model:**
 This model emphasizes usability, performance, and supportability, which are essential for a user-facing system. It ensures the agent is easy to use, fast, and maintainable.

Conclusion:

By combining McCall's and FURPS models, we cover both the internal quality of the system and the user experience. This ensures the voice agent is reliable, efficient, user-friendly, and easy to maintain, making it practical for real-world business adoption.